

Deep Learning Architectures for Tomato Leaf Disease Segmentation: A Comparative Analysis

Joy Chakra Bortty

Department of Computer Science
Westcliff University, Irvine, California,
USA
chakraborttyjoy9@gmail.com

Gouri Shankar Chakraborty

Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, Punjab, India
shankarchakraborty140@gmail.com

Salil Batra

Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, Punjab, India
salilbatra40@gmail.com

Sourav Das Polok

Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, Punjab, India
sourav.polok13@gmail.com

Joy Das

Department of Pharmacology, School
of Pharmaceutical Sciences, Lovely
Professional University, Phagwara,
Punjab
joydas5852@gmail.com

Isha

Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, Punjab, India
sisha0486@gmail.com

Abstract— The growing threat of plant diseases has raised significant concerns for the agricultural sector, particularly affecting vital crops like tomatoes. These diseases, caused by various pathogens such as bacteria, viruses, and fungi, pose a substantial risk to food security by reducing crop yields. Recent years have seen an alarming rise in the occurrence of tomato leaf diseases, largely attributed to environmental changes, leading to the destruction of a significant portion of tomato harvests in many regions. The need for timely and accurate diagnosis of these diseases is now more pressing than ever, as traditional manual methods have proven inefficient and labor-intensive. Advances in artificial intelligence (AI) and deep learning (DL) technologies offer a promising solution to this challenge. Convolutional Neural Networks (CNN) has been particularly effective in automating the identification and segmentation of diseased regions on tomato leaves. By analyzing large datasets of leaf images, these models are capable of detecting disease patterns with high precision. Among the various DL models, UNet++ has demonstrated superior performance, boasting an accuracy of 98.67% in the segmentation of infected areas. Its innovative architectural design, featuring enhanced skip connections and deep supervision, allows for more detailed analysis of complex leaf patterns. Other models such as FCN, SegNet, and Mask R-CNN also contribute to this field, but UNet++ remains the most reliable for high-accuracy tasks. By facilitating early detection and precise treatment, DL models have the potential to significantly reduce crop losses, thereby enhancing productivity and ensuring more stable food supplies in the future.

Keywords—Deep learning, Segmentation, CNN, Tomato Leaf Disease, FPN, FCN, UNet++, Neural Network..

I. INTRODUCTION

Humans have supported large populations for years through farming as their main source of livelihood [1]. However, plant diseases are currently among the most significant threats that farmers must deal with as they have the capacity to significantly reduce agricultural production [2]. Approximately 16 percent of global crop losses occur each year as a result of these diseases [3]. Up to half of some crops' potential harvests, including soybeans and wheat, are being wiped off by pests and diseases [4]. Farmers face challenges in cultivating healthy plants due to the presence of pathogens, including bacteria, viruses, and fungi [5]. Fungi, bacteria, and viruses are all pathogens that affect crop health, including

tomatoes, which are a cornerstone of global meals in both fresh and processed forms such as sauces and ketchup [6]. Tomatoes, one of the most consumed fruits, are a food with significant worldwide importance due to their high annual consumption rates [7]. Over the past few years, tomato leaf diseases have become extremely common, especially from 2020 to 2024. Research and publications have revealed that during this period the amount of bacterial, viral, and fungal illnesses in tomato crop has undergone an increase [7], [8]. Sometimes the diseases have attacked more than 20–30% of tomato plants in some tomato crop regions, causing significant result losses in yield and quality [9], [10]. It seems likely that the primary culprit for this increasing rate of illness prevalence is changes in the climate which create an environment conducive for transmitting infections better [11]. Because tomato plants are extremely susceptible to infectious disease caused by bacterial, viral, and fungal pests, early disease detection technologies have become essential [12], [13]. Farmers are therefore more concerned than ever with the early identification and treatment of tomato leaf diseases [14]. In recent years, artificial intelligence (AI) and deep learning (DL) technologies have been widely used in agriculture to improve plant disease diagnosis [15]. Manual inspection is traditional, time consuming, error prone and labour intensive [16]. Therefore, as a solution, Convolutional Neural Networks (CNN), an amazing application of deep learning, can detect and segment disease on tomato leaves [17]. These AI algorithms can evaluate the leaf images and then analyze enormous pools of leaf images for patterns connected to illnesses, providing very exact diagnoses [18]. By better image segmentation techniques, AI models can be more focused on areas of diseased leaf and therefore performs better [19]. With the increasing risk of tomato leaf disease in recent times, DL techniques for precise identification and separation of disease infected regions offers a viable approach for strong disease management and crop health [20].

In this work, following objectives have been achieved:

1. A thorough literature study has been done where existing techniques and limitations related to Tomato leaf segmentation are covered.

2. Six different DL based models were trained and applied for leaf image segmentation, among which the best model has been selected based on the performance and efficiency for developing the final framework.
3. A comparative analysis has been done where performance of the proposed work has been analyzed and compared with state-of-the-art techniques.

In Sec II, review of literature is done where several existing works have been thoroughly reviewed. In Sec III, proposed methodology has been presented. Sec IV presents the results of the proposed work and discussed based on the comparison with existing state-of-the-art techniques. Conclusion has been drawn in Sec IV. To support the work, a list of references that have been used in this work, is duly attached at the last.

II. LITERATURE REVIEW

In recent years, deep learning methods have impressive impact upon detecting and segmenting tomato leaf diseases indicating the advancement of several studies. Shrivastav et al. (2023) highlight the immensely important role that agriculture has in sustaining human life, including dealing with the details of finding diseases of tomatoes that reduce crop production rates. According to their review, swarm optimization techniques together with clustering based segmentation have the potential to increase detection accuracy [17].

To combat post-harvest spoilage, Reddy et al (2023) use YOLOv4 and YOLOv5 models with impressive classification accuracies of 95.9 and 99.9, accurately classifying fresh produce, including tomatoes [15]. In this field, Wu et al. (2022) take use of EfficientNetV2 and YOLOv5 to train and predict the ten most common tomato leaf diseases, and show improved efficiency and detection accuracy by a deepened YOLOv5 network [21]. The importance of early disease detection is further emphasized by Gangwar et al. (2023), who develop a customized U-Net model, achieving 98.12% accuracy in automated tomato disease detection, thereby highlighting its potential to mitigate crop losses [19].

Ba et al. (2024) introduce an automated classification system using the YOLO algorithm that achieves around 98% accuracy in classifying tomatoes, effectively addressing inefficiencies in traditional manual labor [16]. In a fascinating approach to extremely early disease diagnosis, Brucal et al. (2023) show that YOLOv8 can detect nine tomato leaf illnesses with a mean Average Precision of 98.9% on the same [22]. In Yan et al. (2023), an innovative Si-YOLO model is presented to improve cherry tomato detection in complex greenhouse environments with 95.3 accuracy [18]. Yonggui et al. (2024) boost YOLOv8's detection potential by building on the sophisticated structures and loss functions in order to attain better performance indicators [23]. Similarly, V et al. (2024) propose a hybrid model based on BiLSTM and CNN together with MBIRCH algorithm for tomato disease segmentation and classification [24]. By augmenting YOLOv5s with a Swin transformer attention mechanism, You et al. (2023) show that they effectively tackle the design of efficient robotic systems for tomato harvesting that can achieve both higher accuracy and generalizability [25]. Integrating YOLOv8 for real time

detection and classification of pests and diseases of tomato crop, K et al. (2024) suggest an innovative way to manage greenhouse and prove it is superior to YOLOv5 [26]. Ahmed et al. (2023) report that "Agrobot," an artificial intelligence (AI) driven robotic system for real-time plant disease and pest detection in Sudan, focuses on tomato crops, effectively addresses the issue of quick threat identification [20]. Due to the inefficiencies of traditional manual picking techniques, Guo et al. (2023) suggest an improved YOLOv7 based detection model for robotic picking [27].

Wilson et al. (2023) use the autonomous tomato-harvesting robot based on Turtlebot 2i and YOLOv3, embellished with image pre-processing technology to decrease empty detections, though occlusion remains a problem. This reduces the number of empty detections, although there is still room for improvement [28]. The YOLOv8l-Seg model developed by Uygun et al. (2024) outperformed previous versions in their instance segmentation deep learning study, which aimed to detect *Tuta absoluta* pest damage on tomato plants early [29]. In a Hybrid-DSCNN model for early-stage diagnosis of biotic stress on tomato leaves, Kaur et al. (2024) reported accuracies of 98.24% [30]. In a comprehensive survey, Kumar et. al (2023), presented a massive survey of deep learning techniques used to weed identification and leaf image segmentation which are fast becoming automated processes that were initially botanist's work [31].

Reflecting the challenges in fruit detection caused by occlusions and variation of lighting conditions, Cardellicchio et al. (2023) use transfer learning from YOLOv5 and YOLOv8 models and shows how fruit detection gives high classification accuracy in different farming conditions [32]. Shi et al. (2024) finally present an improved YOLOv5s algorithm for tomato leaf disease detection in complex environments mAP0.5: 93.3% [33]; Liu et al. (2024) propose a YOLOv8 based algorithm on YOLOv8 combined with multi-head self-attention and deformable convolution which improved detection accuracy and speed, and then benefit the prospect of lightweight intelligent terminal application in agricultural [34]. Looking together these studies provide a promising pathway towards deep learning revolutionized technologies in tomato leaf disease detection and segmentation, which would provide better agriculture practices. TABLE I shows the summary of the review of literature.

TABLE I. Summary of Literature Review

Authors	Model Used	Total Images	Dataset	Methodology	Focus Area	Limitations
Kaur et al. (2024) [30]	Hybrid-DSCNN	1004	Plant Village dataset	DL segmentation	Early-stage diagnosis of plant diseases	Potential over-fitting issues
Reddy et al. (2023) [15]	YOLOv4, YOLOv5	1200	Fresh and Stale Images Dataset (Kaggle)	Classification	Classification of fruits and vegetables	Limited scope of classification, focusing only on six types of fruits and vegetables, which may not represent agricultural diversity.
Gangwar et al. (2023) [19]	Customized U-Net	22930	Plant Village Dataset	Image segmentation	Automated disease detection in tomatoes	Limited generalizability
Ba et al. (2024) [16]	YOLO	2500	Tomato dataset	Automated classification	Post-harvest classification	Relies on a limited set of features (size, color, shape) for classification, which may miss other important quality factors.
Brucal et al. (2023) [22]	YOLOv8	1906	Custom Dataset	Detection	Disease detection in tomato production	High precision rates may not translate to real-world conditions where environmental factors affect detection accuracy.
Shi et al. (2024) [33]	YOLOv5s	6430	Plant Village dataset	Improved detection algorithm	Tomato leaf disease detection	May not generalize across datasets
Uygun et al. (2024) [29]	YOLOv8-Seg	800	Custom Dataset	Instance segmentation	Early detection of pest damage	Limited dataset size
You et al. (2023) [25]	YOLOv5-SWINL	1200	Public Datasets, Internet Collection and Field Shooting	Efficient robot picking by proper tomato identification	Enhanced Identification and Accurate Position	Performance under varying conditions
Wu et al. (2022) [21]	EfficientNetV2, YOLOv5	1600	Custom labeled dataset	Identification and Detection	Identification and Detection of 10 common tomato leaf diseases	Relatively small dataset size may affect generalizability of results; model performance might vary under different lighting conditions, impacting detection accuracy.

III. PROPOSED METHODOLOGY

A. Dataset Description

To train a model, the first step is to collect data. For this purpose, data can be gathered from Kaggle, which provides authentic datasets. One such dataset is the "Tomato Leaf Segmentation" dataset, which is divided into two parts: leaf images and mask images. The mask images contain the affected parts of the corresponding leaf images. The total number of images is 6,360, with 3,180 leaf images and 3,180 mask images. This dataset is useful for training a model to effectively segment tomato leaf diseases. The dataset can be accessed at the following link: <https://www.kaggle.com/datasets/souravdaspolok/tomato-leaf-segmented-images/data.dataset>.

B. Preprocessing

Improving the quality of the dataset through data preprocessing is important for better tomato leaf disease

segmentation. To ensure consistency, every image is resized to 256x256x256 pixels. Utilize augmentation methods on the suggested dataset, such as rotation, flipping, scaling, elastic deformation, and intensity shifts. Total 3180 images with 3100 mask images were preprocessed and split into training and validation part where 2800 samples were used for training and 380 samples were used for testing. Model accuracy, reduce loss and dataset variability is increased through augmentation process.

C. Feature selection, extraction and fusion technique

The subsequent phase is to identify the most relevant features from the tomato leaf disease images, which assists in the identification of effected areas after the dataset has been collected, combined, and preprocessed. Feature selection is the term used to describe this method. Feature extraction enhances the efficacy of the model and reduces the complexity of the data. Meaningful information, such as the creation of mask images, can be obtained following feature

extraction. This step is also included in the feature extraction process. To extract features and characterize the characteristics of the affected area, a variety of techniques are employed, including wavelet transforms, deep learning models, and image analysis. After the features are extracted, they are combined to generate a more comprehensive dataset, a process labeled "fusion." The model's ability to detect malignancies is improved by fusion.

D. Segmentation

The process of identifying a portion of an image using features that the training model has learned is known as segmentation. Although there are many other kinds of segmentation techniques, medical picture segmentation is essential to this job. In our work, we use this segmentation technique to determine the area affected from tomato leaf disease images. Many techniques, including Fully Convolutional Networks (FCN), FPN, U-Net, UNet++, SegNet, DeepLab, and Mask R-CNN, are frequently used for segmentation problems with Convolutional Neural Networks (CNNs).

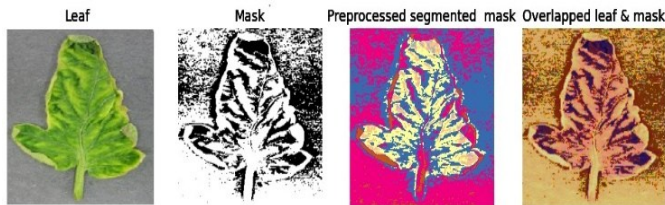


Fig 1. Difference between segmented images and original images

E. Architecture of the CNN models used in the work

Convolution Neural Network (CNN) is an updated variant of artificial neural networks (ANN) that is employed to extract features from visual datasets that resemble grids. Input, convolutional, pooling, and fully connected layers comprise CNN. The input layer is responsible for providing the model with the input, which is a sequence of images. Convolutional layers are employed to derive features from this input image. The kernel contains matrix shapes of 2×2 , 3×3 , or 5×5 . This kernel partitions the input images according to their features. The CNN activation layer offers a variety of activation functions, including ReLU, Leaky ReLU, and Sigmoid. It facilitates the extraction of meaningful features and the modeling of non-linear relationships. The features are converted to the primary function in the pooling layer, which reduces the size of the feature. The two most prevalent varieties of pooling layers are average pooling and maximum pooling. The pooling layer is responsible for providing input to the fully connected layer and performing the segmentation duties.

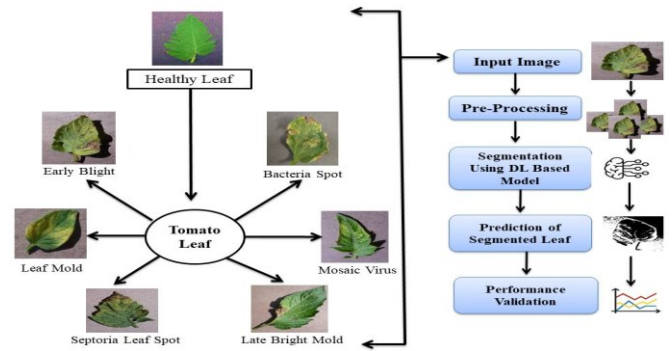


Fig 2. Flow diagram of the proposed method

a) Feature Pyramid Network (FPN) architecture

A Feature Pyramid Network (FPN) functions as a feature extractor, converting a single-scale image into a multi-scale object detection model. It is a neural network architecture design. To transform single-scale images into multi-scale images with features, FPN is a hybrid combination of multiple object detection models, including Mask R-CNN and RetinaNet.

b) Fully Convolutional Network (FCN) architecture

A fully convolutional network (FCN) architecture that is specifically designed for semantic segmentation. It accepts images of any size as input and processes them using multiple convolutional layers to generate a pixel-by-pixel prediction map. Every pixel is designated a class, and it is responsible for the segmentation process. Many advanced segmentation architectures were created using the FCN architecture.

c) UNet architecture

UNet is employed in the segmentation of biomedical images. Additionally, it is employed to segment medical images. The U-Net architecture is characterized by a "U" configuration and comprises two paths: the contracting path and the expansive path. The contracting path is used to identify the feature of the input images and map it. Improve the expansive path feature of input images.

d) UNet++ architecture

UNet++ is an improved variant of UNet. In this algorithm, densely connected skip pathways and nested are introduced, which involve the addition of the encoder and decoder path to convolutional layers, thereby modifying the transfer information. It is beneficial to transition between numerous phases that have more integrated features.

e) Mask R-CNN architecture

A deep learning algorithm was used to develop the advanced segmentation algorithm known as Mask R-CNN. A pixel-level mask of each object is used by this algorithm to identify objects in an image. This algorithm operates in two phases. The Region Proposal Network (RPN) identifies the portion of the images that contain the regions of interest. In other stages, the region was classified and a pixel-wise mask was generated for segmentation.

f) SegNet

SegNet is employed for semantic segmentation. It created classifications for each pixel in an image. It includes encoder and decoder structures. The encoder extracts the feature from the convolution layer, such as VGG16. The decoder works to preserve the location of the encoder's pooling stages in order

to restore the image during upsampling. SegNet is more memory-efficient and faster in real-time operations.

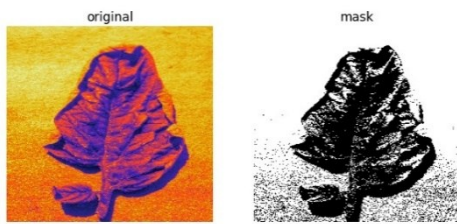


Fig 3. Mask images of Tomato Leaf Disease

IV. RESULTS AND DISCUSSION

This section addresses the outcomes and real-time performance evaluation of the proposed framework. Fig 4. mentions the accuracy of every model address int the proposed work where UNet++ provides the best accuracy from other model with 98.67% accuracy.

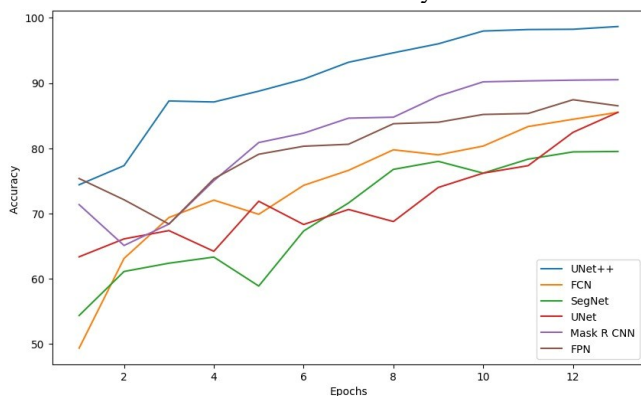


Fig 4. Performance analysis of various models used in proposed framework

Table II. presents the performance statics of all models employed in this proposed work. After analyzing the table, it can be concluded that UNet++ provides the best performance compared to other models in terms od accuracy, IoU, dice Cof and loss.

TABLE II. SHOWS THE PERFORMANCE OF DIFFERENT MODELS

Models	Accuracy	IoU	Dice Coef	Loss
UNet++	98.67	95.51	96.38	0.049
FCN	85.51	83.18	81.10	0.156
SegNet	79.53	74.62	71.65	0.219
UNet	85.62	76.64	74.89	0.154
Mask R CNN	90.06	88.64	89.96	0.104
FPN	86.24	77.95	83.61	0.148

TABLE III. displays the performance information where mention the epoch wise accuracy, IoU,dice coef and loss of UNet ++ and highest scores of accuracy, IoU and dice coef were achieve in 13th epoch with minimum loss.

TABLE III. EPOCH WISE PERFORMANCE OF UNET++

Epoch	Accuracy	Dice Coef	Loss	IoU
1	74.41	0.6548	0.39	0.8236
2	77.34	0.6833	0.37	0.8510

3	87.26	0.7424	0.38	0.8638
4	87.10	0.8820	0.38	0.8832
5	88.76	0.9004	0.31	0.9088
6	90.60	0.9246	0.22	0.9031
7	90.60	0.9246	0.16	0.9061
8	93.19	0.9592	0.15	0.9171
9	94.64	0.9571	0.09	0.9299
10	97.98	0.9530	0.08	0.9318
11	98.21	0.9679	0.08	0.9433
12	98.25	0.9614	0.05	0.9544
13	98.67	0.9638	0.04	0.9551

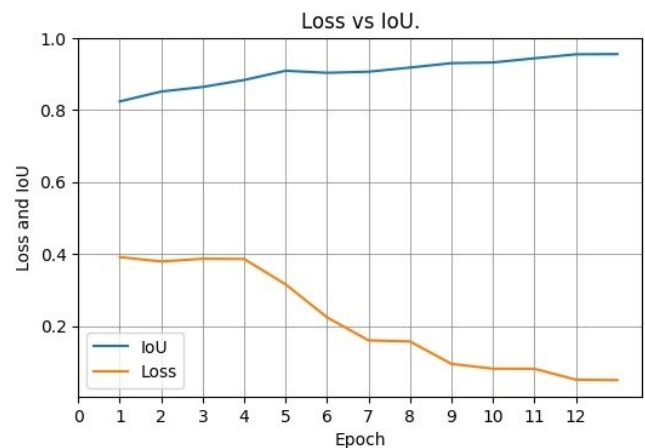


Fig 5. Epoch wise Loss and IoU Graph for UNet++

Fig 5.shows the IoU and lose performance of UNet++ with epoch varitions. IoU was low and loss was high in the 1st epochs. After increasing the epochs the IoU was increasing and loss went downward. At the 13th epoch, the highest IoU was achieved of .9638 with minimum loss of less than .04.

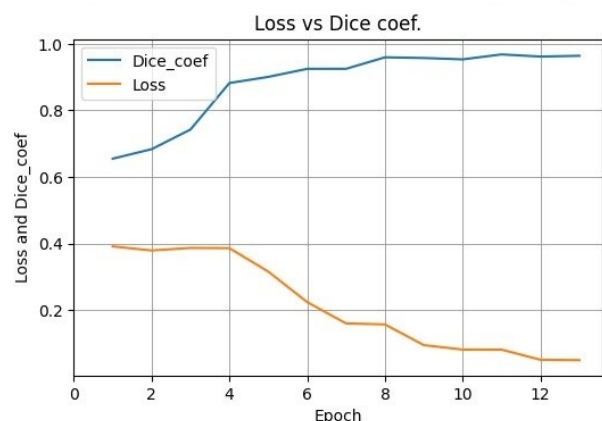


Fig 6. Epoch wise Loss and Dice_coef graph for UNet++

V. CONCLUSION

In fig 6.shows the loss and dice coef. wise performance of UNet++ with each epoch. In the beginning, the loss was high and the dice coef. was low. When the epoch count was increased the performance dice coef. values was increased and loss was gone to low. This model provided the best dice coef. values at 13th epoch with minimum loss less than .04. At 13th epoch the dice coef. value was 96.38%.

TABLE IV. COMPARATIVE PERFORMANCE ASSESSMENT OF PROPOSED METHOD AND EXISTING APPROACHES

Authors	Model Used	Accuracy	Total Images
K et al. (2024) [26]	YOLOv8	82.4%	10269
Wu et al. (2022) [21]	EfficientNetV2, YOLOv5	83.9%	1600
You et al. (2023) [25]	YOLOv5-SWINL	91.622%	1200
Shi et al. (2024) [33]	YOLOv5s	93.3%	6430
Uygun et al. (2024) [29]	YOLOv8-Seg	93.5%	800
Proposed Method	UNet, UNet++, FPN, FCN, Mask R-CNN, SegNet	98.67%	3180

TABLE IV. described the compression between existing work to proposed work. Six segmentation algorithm used in this work and between them UNet++ provides good accuracy of 98.67% using 3180 images of tomato leaf.



Fig 7.Real time output of Tomato Leaf Diseases Segmentation

Fig 7.display the output came from the proposed framework. The first image mentioned as the original is a preprocessed image from the dataset. The second one is mask image that mention in dataset as the y label. The final one is the predicted mask images generated by the proposed model, which is substantially identical to the real mask image justify the performance of 98.67% accuracy.

In conclusion, DL methods have become essential tools for detecting and segmenting tomato leaf diseases, offering significant advantages over traditional approaches. However, manual methods are not efficient and reliable for identifying plant diseases that are frequently spreading with time. DL models are a better approach to early disease diagnosis adhering to maximum preciseness which is paramount for protecting crop yield and thus the food security. Among the DL models, UNet++ stands out as the top performer, achieving an impressive accuracy of 98.67%. Its advanced architecture, incorporating redesigned skip connections and deep supervision, allows it to capture complex patterns in leaf images, making it highly effective in precisely segmenting diseased areas. This capacity makes the UNet++ model particularly suitable for tasks categorically analyzing influenced regions and distinguishes it from other models. Other models which have the relevance in tomato leaf disease detection include FCN, SegNet, UNet, Mask R CNN and FPN. For instance, SegNet is specifically designed for real-time applications since it is little and FPN as the name implies is good for feature pyramid. However, despite their efforts all of these models are a bit off the mark from what UNet++ delivers. In the future developments of the DL technology, the segmentation methods together with the feature extraction techniques will also enhance the efficiency of the current models. This will enable farmers to detect the diseases affecting the plants, and avoid loss of crop through diseases, especially given the escalating problems facing agriculture.

REFERENCES

- [1] I. S. Bisht, J. C. Rana, and S. P. Ahlawat, "The future of smallholder farming in India: Some sustainability considerations," *Sustain.*, vol. 12, no. 9, pp. 1–25, 2020, doi: 10.3390/su12093751.
- [2] J. B. Ristaino et al., "The persistent threat of emerging plant disease pandemics to global food security," *Proc. Natl. Acad. Sci. U. S. A.*, vol. 118, no. 23, pp. 1–9, 2021, doi: 10.1073/pnas.2022391118.
- [3] D. S. Mueller et al., "Corn yield loss estimates due to diseases in the United States and Ontario, Canada, from 2016 to 2019," *Plant Heal. Prog.*, vol. 21, no. 4, pp. 238–247, 2020, doi: 10.1094/PHP-05-20-0038-RS.
- [4] O. T. Toomer et al., "Current Agronomic Practices, Harvest & Post-Harvest Processing of Soybeans (*Glycine max*)—A Review †," *Agronomy*, vol. 13, p. 427, 2023, [Online]. Available: <https://doi.org/10.3390/agronomy13020427>
- [5] M. Jeger et al., "Global challenges facing plant pathology: multidisciplinary approaches to meet the food security and environmental challenges in the mid-twenty-first century," *CABI Agric. Biosci.*, vol. 2, no. 1, pp. 1–18, 2021, doi: 10.1186/s43170-021-00042-x.
- [6] S. Karanth, S. Feng, D. Patra, and A. K. Pradhan, "Linking microbial contamination to food spoilage and food waste: the role of smart packaging, spoilage risk assessments, and date labeling," *Front. Microbiol.*, vol. 14, no. June, pp. 1–17, 2023, doi: 10.3389/fmicb.2023.1198124.
- [7] E. J. Collins, C. Bowyer, A. Tsouza, and M. Chopra, "Tomatoes: An Extensive Review of the Associated Health Their Cultivation," *Biology (Basel)*, vol. 11, p. 239, 2022.
- [8] A. Saeed, A. A. Abdel-Aziz, A. Mossad, M. A. Abdelhamid, A. Y. Alkhaled, and M. Mayhoub, "Smart Detection of Tomato Leaf Diseases Using Transfer Learning-Based Convolutional Neural Networks," *Agric.*, vol. 13, no. 1, 2023, doi: 10.3390/agriculture13010139.
- [9] S. Panno et al., "A review of the most common and economically important diseases that undermine the cultivation of tomato crop in the mediterranean basin," *Agronomy*, vol. 11, no. 11, pp. 1–45, 2021, doi: 10.3390/agronomy11112188.
- [10] A. K. Pandey, A. Kumar, K. Dinesh, R. Varshney, and P. Dutta, "The hunt for beneficial fungi for tomato crop improvement – Advantages and perspectives," *Plant Stress*, vol. 6, no. June, p. 100110, 2022, doi: 10.1016/j.stress.2022.100110.

- [11] V. GHOSH, P.K. Jeer, MALLIKARJUNA, SIVALINGAM, P.N, PARAMESHWAR, B., SINGH, H.K, Choudhary, K. K. Kumar, B. H. I. M. E. S. H. W. A. R. I. Sahu, S. K. Muthappa, and A. D. A. Das, "Agronomic innovations in biotic stress management and its combined effect with abiotic stresses in crop production," *Indian J. Agron.*, vol. 66, no. 5, pp. 237–257, 2021.
- [12] J. Liu and X. Wang, "Tomato Diseases and Pests Detection Based on Improved Yolo V3 Convolutional Neural Network," *Front. Plant Sci.*, vol. 11, no. June, pp. 1–12, 2020, doi: 10.3389/fpls.2020.00898.
- [13] D. M. S. K. Dr. V. Anantha Natarajan, Ms. Macha Babitha, "Detection of disease in tomato plant using Deep Learning Techniques," *Int. J. Mod. Agric.*, vol. 9, no. 4, pp. 2305–7246, 2020, doi: 10.47059/alinteri/v36i2/ajas21118.
- [14] N. K. Trivedi et al., "Early detection and classification of tomato leaf disease using high-performance deep neural network," *Sensors*, vol. 21, no. 23, 2021, doi: 10.3390/s21237987.
- [15] M. S. S. A. Reddy and N. Aishwarya, "A Deep Learning Approach to Identify Fresh and Stale Fruits and Vegetables with YOLO," *IEEE Int. Conf. Adv. Electron. Commun. Comput. Intell. Inf. Syst. ICAECIS 2023 - Proc.*, pp. 606–610, 2023, doi: 10.1109/ICAECIS58353.2023.10170163.
- [16] Q. H. Do Ba et al., "Developed a Tomato-Condition Classification System using Image Processing and YOLO Technology," *Proc. 2024 9th Int. Conf. Integr. Circuits, Des. Verif. ICDV 2024*, no. Icdv, pp. 108–113, 2024, doi: 10.1109/ICDV61346.2024.10616990.
- [17] S. Shrivastav, V. Jindal, R. Eswarawaka, and D. Ray, "A Comprehensive Review and Discussion of Segmentation based Tomato Plant Disease Detection Approaches," *2023 IEEE Int. Conf. Res. Methodol. Knowl. Manag. Artif. Intell. Telecommun. Eng. RMKMATE 2023*, pp. 1–7, 2023, doi: 10.1109/RMKMATE59243.2023.10369911.
- [18] Y. Yan, J. Zhang, Z. Bi, and P. Wang, "Identification and Location Method of Cherry Tomato Picking Point Based on Si-YOLO," *Proc. 13th IEEE Int. Conf. CYBER Technol. Autom. Control. Intell. Syst. CYBER 2023*, pp. 373–378, 2023, doi: 10.1109/CYBER59472.2023.10256630.
- [19] A. Gangwar, G. Rani, V. P. S. Dhaka, and Sonam, "Detecting Tomato Crop Diseases with AI: Leaf Segmentation and Analysis," *7th Int. Conf. Trends Electron. Informatics, ICOEI 2023 - Proc.*, no. Icoei, pp. 902–907, 2023, doi: 10.1109/ICOEI56765.2023.10125891.
- [20] A. J. O. Ahmed, A. Babiker, A. Elhag, and M. Drar, "Real-Time Agricultural Monitoring with Agrobot: A Raspberry Pi and YOLO Based Solution," *ICCA 2023 - 2023 5th Int. Conf. Comput. Appl. Proc.*, pp. 1–5, 2023, doi: 10.1109/ICCA59364.2023.10401368.
- [21] X. H. Wu, X. Li, S. Kong, Y. Zhao, and L. Peng, "Application of EfficientNetV2 and YoloV5 for tomato leaf disease identification," *Proc. - 2022 Asia Conf. Algorithms, Comput. Mach. Learn. CACML 2022*, pp. 150–158, 2022, doi: 10.1109/CACML55074.2022.00033.
- [22] S. G. E. Brucal, L. C. M. De Jesus, S. R. Peruda, L. A. Samaniego, and E. D. Yong, "Development of Tomato Leaf Disease Detection using YoloV8 Model via RoboFlow 2.0," *GCCE 2023 - 2023 IEEE 12th Glob. Conf. Consum. Electron.*, pp. 692–694, 2023, doi: 10.1109/GCCE59613.2023.10315251.
- [23] W. Yonggui and H. Jing, "Improved Algorithm for Tomato Disease Detection Based on YOLOv8," *2024 IEEE 4th Int. Conf. Electron. Technol. Commun. Information, ICETCI 2024*, pp. 230–234, 2024, doi: 10.1109/ICETCI61221.2024.10594250.
- [24] V. Jayanthi and M. Kanchana, "Modified BIRCH and Hybrid Deep Framework for Tomato Leaf Disease Segmentation and Classification," *Proc. - 2024 5th Int. Conf. Intell. Commun. Technol. Virtual Mob. Networks, ICICV 2024*, no. Iciev, pp. 149–160, 2024, doi: 10.1109/ICICV62344.2024.00030.
- [25] S. You, J. Bian, and C. Wang, "Optimization of an improved YOLOv5s algorithm for tomato identification," *2023 3rd Int. Conf. Electron. Inf. Eng. Comput. EIECT 2023*, pp. 504–507, 2023, doi: 10.1109/EIECT60552.2023.10442944.
- [26] L. K. Arjun, Shine, "Pest Detection and Disease Categorization in Tomato Crops using YOLOv8," *2024 IEEE Recent Adv. Intell. Comput. Syst.*, pp. 1–6, doi: 10.1109/RAICS61201.2024.10690151.
- [27] J. Guo et al., "Revolutionizing Agriculture: Real-Time Ripe Tomato Detection With the Enhanced Tomato-YOLOv7 System," *IEEE Access*, vol. 11, no. November, pp. 133086–133098, 2023, doi: 10.1109/ACCESS.2023.3336562.
- [28] A. Wilson, A. Bisht, and L. Milkowski, "Robotic Tomato Detection using YOLOv3 with Pre- and Post-Processing: Evaluation through Gazebo Simulation and Real-World Laboratory Testing," *Proc. - 2023 17th IEEE Int. Conf. Robot. Comput. IRC 2023*, pp. 255–259, 2023, doi: 10.1109/IRC59093.2023.00048.
- [29] T. Uygun and M. M. Ozguven, "Determination of tomato leafminer: Tuta absoluta (Meyrick) (Lepidoptera: Gelechiidae) damage on tomato using deep learning instance segmentation method," *Eur. Food Res. Technol.*, vol. 250, no. 6, pp. 1837–1852, 2024, doi: 10.1007/s00217-024-04516-w.
- [30] P. Kaur, S. Harnal, V. Gautam, M. P. Singh, and S. P. Singh, "Performance analysis of segmentation models to detect leaf diseases in tomato plant," *Multimed. Tools Appl.*, vol. 83, no. 6, pp. 16019–16043, 2024, doi: 10.1007/s11042-023-16238-4.
- [31] A. Kumar and S. Sachar, *Deep Learning Techniques in Leaf Image Segmentation and Leaf Species Classification: A Survey*, vol. 133, no. 4. Springer US, 2023. doi: 10.1007/s11277-024-10873-2.
- [32] A. Cardellicchio, V. Reno, R. P. Devanna, R. Marani, and A. Milella, "Tomato detection in challenging scenarios using YOLO-based single stage detectors," *2023 IEEE Int. Work. Metrol. Agric. For. MetroAgriFor 2023 - Proc.*, pp. 54–59, 2023, doi: 10.1109/MetroAgriFor58484.2023.10424305.
- [33] Z. Shi, J. Liu, X. Qin, and J. Cao, "Tomato Leaf Disease Detection Algorithm Based on YOLOv5s," *Chinese Control Conf. CCC*, pp. 8703–8708, 2024, doi: 10.23919/CCC63176.2024.10662742.
- [34] Y. Z. Xiangyu Liu, Haotian Lei, Yan Zhou, Jia ming Feng, Guo Niu, "Tomato leaf disease detection based on improved YOLOv8," *2024 6th Int. Conf. Internet Things, Autom. Artif. Intell.*, vol. 13, no. 3, pp. 145–150, 2024, doi: 10.1109/IoTAAI62601.2024.10692846.